Inside the Frontier: Ownership Change and Organizational Inertia in Nursing Home Production*

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2026-05-31

Abstract

Health care providers may fail to continuously update clinical and managerial routines, leaving them inside their feasible production frontier. We study this possibility in U.S. nursing homes, where ownership changes create moments of organizational disruption and reoptimization. Using CMS administrative panel data from 2017-2024, we identify facility-level ownership transfers and estimate staffing and quality responses in a staggered-adoption event-study design. Following ownership changes, facilities reduce registered nurse and nurse aide hours, while licensed practical-nurse hours remain largely unchanged. Routine-sensitive quality measures, including catheter use, psychotropic medication use, and urinary tract infections, improve. By contrast, staffing-intensive resident outcomes such as pressure injuries, falls, weight loss, and decline in physical functioning show little evidence of broad deterioration. These patterns are consistent with ownership transitions revealing inside-frontier inefficiency: new owners do not necessarily possess superior care technology, but may be more willing to revise stale routines, reduce organizational slack, and move facilities closer to efficient operating practices. The findings suggest that governance disruptions can reshape health care production through routine replacement, with ambiguous implications for resident welfare when input reductions and process-quality improvements occur simultaneously.

Keywords: nursing homes, ownership change, staffing, difference-in-differences

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JEL Codes: I11, I18, J22, L22, L33

1 Introduction

Ownership changes are increasingly common in healthcare markets and have become a growing concern for policymakers. In U.S. Nursing homes, Skilled nursing facilities (SNFs) can change ownership through acquisitions, sales, and corporate reorganizations, and these changes are frequently talked about because of their possible implications to the quality of care (cite here). Between 2016 and 2021 more than 3,200 SNFs were sold in the United States (Prusynski et al., 2023), and this number will continue to increase well into the future (cite here). Nursing homes serve medically vulnerable residents who depend on staff for both oversight and daily living assistance, even small changes in staffing could translate into significant differences in care. We study nurse staffing as a central input to the nursing home care production, through which ownership changes may affect care delivery. Specifically we ask, how do ownership changes affect nurse staffing levels in U.S. nursing homes, and do effects differ in chains vs non-chains and before versus during the COVID-19 pandemic?

A large body of industrial organization literature studies mergers and acquisitions that reshape market structure by changing the set of competing providers in a market, raising concerns about market power, prices, wages, and competition. In healthcare, this work often evaluates consolidation through hospital, physician, or insurer mergers that increase or decrease concentration and can affect negotiated prices and input costs, making antitrust scrutiny central to the policy debate. The ownership changes we study are different. Rather than eliminating a competitor or combining two providers, most events in our data reflect a transfer of control over an existing nursing home: the facility remains in the same physical location, continues operating under the same certification framework, and continues serving a local resident population facing the same outside options and regulatory environment. As a result, these ownership changes do not mechanically alter the number of providers in the market or the underlying market structure. This distinction matters, because in settings where prices are largely fixed in the short run and market structure is unchanged, the primary channels through which ownership change can affect welfare are internal production decisions such as management practices, staffing, and other operational choices, rather than conventional market-power effects emphasized in classic industrial organization literature.

Nursing homes provide a particularly important setting in which to study the non-price consequences of ownership changes. Shortcomings in nursing home quality has been a long standing concern, and the policy debate has repeatedly emphasized incentives and constraints that shape providers' quality choices (Hackmann, 2019). Nursing home markets also exhibit severe frictions: residents and families place substantial weight on proximity, making facility decisions central to the care experienced by local populations (Gupta et al., 2021). These frictions imply that ownership changes can affect resident welfare through internal production decisions even when the facility remains open and continues to serve the same people facing the same regulations. Staffing is an especially natural outcome to study because it is both highly important for quality and one of the few outcomes that can adjust relatively quickly in response to changes in ownership.

Nursing labor is the core input in nursing home care production, and staffing is one of the most consequential operation choices facilities make. We distinguish among registered nurses (RNs), licensed practical nurses (LPNs) and certified nursing assistants (CNAs). RNs include registered nurses and directors of nursing and play central supervisory and clinical roles, including assessing residents' conditions, developing and coordinating care plans, overseeing medication-related processes, and supervising other staff. LPNs provide routine clinical services and medication management tasks such as monitoring vital signs, while CNAs deliver the majority of hands-on daily care, assisting with bathing, dressing, toileting, feeding, and mobility (Lin, 2014). Because these staff types differ in training, scope of practice, and cost, ownership changes may affect not only total staffing hours but also the skill mix of care delivered. For example, an owner seeking quick cost containment may reduce total hours or substitute away from higher-cost licensed nurses, while an owner focused on compliance or clinical quality may increase RN oversight or adjust supervision structures.

A large body of evidence indicates that staffing is directly linked to nursing home quality. Early work shows that fewer RN hours and fewer aide hours are associated with more deficiencies, including deficiencies related to quality of care and quality of life (Harrington et al., 2000). Importantly, however, staffing is an endogenous choice made jointly with other inputs and under a set of constraints such as regulation and budgets that complicate causal interpretation of simple staffing and quality correlations.

Our focus on staffing responses to ownership changes is also motivated by literature which shows that financial incentives and ownership structure can have an influence care in nursing homes. Private Equity (PE) ownership of nursing homes has been a rising policy concern, and PE ownership incentives can affect patient outcomes and operational decisions. Gupta et al. (2021) find large increases in short-term mortality associated with PE owned nursing homes and show that staffing composition changes following PE acquisition. They find declines in LPNs and CNAs alongside increases in RN hours, highlighting staffing as a mechanism linking ownership incentives to resident outcomes. While private equity represents one specific ownership form, these findings prove a more general point that is central to this paper: changes in ownership can shift objectives and constraints in ways that reshape internal production decisions, including both staffing levels and skill mix.

Research around how staffing responds to ownership changes is mixed and can be ambiguous. Ownership changes can alter managerial objectives, how owners trade off margins against quality, compliance, and reputation, and can also change the financial environment facing a facility. On one hand, new owners may face stronger incentives to reduce operating costs due to tighter budget targets or financing obligations, making labor cost reductions attractive in a sector where staffing represents a large share of operating expenses. Ownership changes may also introduce short-run disruptions such as turnover in administrators, changes in scheduling and staffing policies, contract renegotiations, and changes in recruitment and retention practices. These channels point toward reductions in staffing or substitution toward lower-cost inputs. On the other hand, ownership changes could increase staffing if new owners place greater weight on quality and compliance, seek to improve performance, or attempt to protect reputation and occupancy in a heavily monitored sector. Public reporting and quality oversight can heighten the incentive to invest in quality and staffing, particularly when quality is noticeable and easily measured (Park and Werner, 2011). In this framework, ownership changes can plausibly generate either quality reducing cost containment or quality improving investment, and the direction may differ by staff type.

This cost–quality tradeoff is highlighted by the fact that nursing home revenues are mostly fixed in the short run. Nursing homes are heavily reliant on public payers, and limited pricing flexibility can constrain the ability to finance staffing investments. Gupta et al. (2021) emphasize this reliance on subsidy and market frictions, and Bowblis and Brunt (2025) give direct evidence that financial resources are tightly linked to staffing: facilities with high Medicaid payer mix have lower staffing levels and staffing expenditures, consistent with the idea that inadequate reimbursement and thin margins limit the ability to hire and retain nursing staff.

More broadly, ownership changes can also affect the cost structure of facilities through channels that do not translate into patient care investment. Holmes (1996) shows that ownership changes are associated with higher capital related costs, plant costs, interest costs, and taxes, without corresponding increases in patient-care expenditures, implying that facility sales may shrink operating margins and increase pressure to cut variable costs such as staffing. Regulation can further shape how facilities adjust: in response to minimum quality standards, facilities may increase staffing but offset the cost through reductions in other inputs, and they may substitute across nurse types in ways that differ across settings (Bowblis and Ghattas, 2017). This institutional setting implies that staffing could rise, fall, or shift in composition following ownership change, making the net effect an empirical question and motivating a design that can detect these ambiguous adjustments.

This paper contributes to several related literatures. First, it adds to the growing body of work on ownership changes and provider performance by focusing on internal production decisions, staffing, rather than solely on market structure or prices. Second, it contributes to the nursing home literature by studying how a major organizational event, an ownership change, effects the central input into care, building on evidence that staffing improves quality (Lin, 2014) and that staffing shortfalls are linked to poor regulatory performance (Harrington et al., 2000). Third, it complements work on financial incentives and quality in nursing homes by explicitly connecting ownership transitions to staffing adjustments in an environment where financial performance can shape quality investment (Park and Werner, 2011) and where Medicaid reliance constrains staffing resources (Bowblis and Brunt, 2025). By examining heterogeneity by chain affiliation and by period relative to COVID-19, the paper sheds light on how organizational structure and shifting labor market conditions mediate staffing responses to ownership change.

To answer the research question, we use Centers for Medicare and Medicaid Services (CMS) date to construct a panel of nursing homes across the 48 contiguous U.S. states from January 2017 through June 2024. We measure staffing outcomes using hours per resident day (HPRD) for RNs, LPNs, CNAs, and total nursing hours. We estimate treatment effects using a staggered adoption difference-in-differences event study design. The results show persistent declines in nursing staffing following ownership changes, concentrated among RNs, CNAs, and total nursing hours, with heterogeneity by chain affiliation and the COVID-19 pandemic.

2 Data and Methods

2.1 Data Sources

We utilize three main sources of data: Nursing Home Compare (NHC) Archive data, Healthcare Cost Report Information System (HCRIS), and Payroll Based Journal (PBJ) data. The NHC Archive contains monthly facility level information on ownership, chain affiliation, certification status, as well as quarterly reported quality measures. Second, we use HCRIS, also known as the Medicare Cost Reports, which contains annual cost reports submitted by every Medicare certified nursing home in the United States. The cost reports contain information about costs, revenue, financial data for each reporting facility, and dates of ownership changes. Lastly, we use the PBJ data to obtain staffing information. The PBJ data contains the resident census and the number of hours worked by various types of NH staff on a daily basis.

2.2 Sample selection

We utilize two samples in our study, the first is our staffing panel which is constructed at the facility-month level and the second is our quality metric panel which is constructed at the facility-quarter level. We construct both samples using the following criteria. First, we limit our samples to the 48 contiguous U.S. states from the period January 2017 to June 2024. Secondly, the NHC Archive data is matched to the HCRIS data on a unique nursing home provider number. Because HCRIS only contains information on freestanding nursing homes, our samples exclude hospital-based nursing homes. Hospital based nursing homes differ from freestanding facilities in their structure and financing. Because they are within hospitals, staffing policies and resource allocation may be jointly determined with hospital operations and respond to hospital shocks and regulations rather than nursing home specific incentives. We therefore exclude hospital based facilities to maintain a more homogeneous set of providers and to ensure ownership changes reflect changes in nursing home management rather than broader hospital system changes. Hospital based nursing homes make up about 12 percent of all nursing homes in the United States.

2.3 Identifying Ownership Changes

No single data source provides a comprehensive record of ownership changes for U.S. nursing homes, so we combine information from HCRIS and the NHC Archive in a two-step identification and verification process to identify ownership changes.

Working on writing this so that it emphasizes this as more of a measurement contribution

First, we use the HCRIS data which contains an indicator for an ownership change, and corresponding date, in each cost report after an ownership change. We use this as our flag for ownership changes during the study period. However, these indicators may capture administrative or legal changes that do not reflect meaningful transfers of control. Thus, to verify these changes, our second step was to validate each potential ownership change in the HCRIS data with the NHC Archive data. The NHC Archive data reports ownership shares for each owner associated with a facility on a monthly basis. One issue with the NHC Archive data is that it reports the direct owners and indirect owners of each nursing home. For example, a limited liability corporation may directly own the nursing home, but that limited liability corporation is owned by different individuals. Therefore, we identified ownership changes to occur among direct and indirect owners using the following criteria:

Using this information, we define a change in ownership when a majority of ownership shares turn over between month $t-1$ and month t. Intuitively, this captures transactions in which control of the facility changes hands, including cases where a new operator acquires most or all shares. Secondly, when ownership percentages are partially or completely missing, we approximate shares by normalizing any reported percentages and, where necessary, assigning equal weights across remaining owners. To reduce false positives from purely legal restructuring, we do not flag a change in ownership if at least 80 percent of the ownership stake remains within the same surname family (e.g., a change from "Smith Holdings LLC" to "Smith Family Partners LP").

Using these verified ownership changes, the analytic sample included nursing homes that either (i) have no ownership change in either source or (ii) have exactly one ownership change in each source with the ownership change date within six months of one another.

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For these Nursing Homes, we define the change in ownership month as the first month in which HCRIS identifies a change. All dates are converted to calendar months, so changes occurring at the beginning or end of a month are treated identically.

2.4 Nursing Staff Levels

The nursing staff types we use in our study are registered nurses (RNs), licensed practical nurses (LPNs), and certified nursing assistants (CNAs). RNs are the most highly trained nursing staff and are responsible for assessment, care planning, medication administration, and supervision of other nursing personnel. LPNs provide basic medical care under the supervision of RNs and physicians, including monitoring vital signs and administering certain treatments. CNAs provide the majority of direct daily care to residents, assisting with activities of daily living. Because these staff types differ in training, scope of practice, and compensation, changes in staffing composition may reflect different organizational responses to ownership changes. We therefore examine each staff category separately, as well as total nurse staffing.

Nursing staff levels are calculated from PBJ data, which report daily staffing hours and resident census. For each facility i, calendar month t, and staff type, we aggregate daily hours and resident days to construct hours per resident day (HPRD):

$$\mathrm{HPRD}_{it} = \frac{\sum\limits_{d \in t} H_{id}}{\sum\limits_{d \in t} R_{id}}, \tag{1}$$

where H_{id} is the number of hours worked by the staff type in facility i on day d, and R_{id} is the number of resident days in facility i on day d. We construct separate HPRD measures for RNs, LPNs, and CNAs, and define total HPRD as the sum of the three type-specific measures. For quality specifications estimated at the facility-quarter level, we construct quarterly staffing measures analogously by summing hours and resident days over all days in the quarter before taking the ratio.

We also construct additional measures of data quality. First, we calculate a monthly PBJ coverage measure, defined as the share of days in the month with complete staffing records. This measure allows us to identify incomplete data in our sample and describe our data completeness. We use this measure for sample construction and descriptive purposes, excluding facilities and observations with low reporting coverage. Secondly, we track reporting gaps, defined as the number of months since a facility's previous PBJ observation, and use this variable in robustness checks that restrict the sample to facilities with more continuous reporting.

Finally, we exclude a small number of observations with implausible staffing levels: months in which RN and LPN HPRD are both zero, total HPRD is below 1.5 or above 12, or CNA HPRD exceeds 5.25. These observations are outliers that likely reflect reporting errors rather than genuine variation in staffing. These exclusion criteria align with the criteria outlined in the CMS NHC Technical Users Guide.

2.5 Quality Metrics

This entire section is new

We measure quality using selected CMS quality measures from the NHC Archive. These measures are reported at the facility-quarter level and are constructed from resident assessments submitted through the Minimum Data Set (MDS). We focus on measures that are plausibly responsive to ownership-related changes in staffing, care routines, medication management, monitoring, and documentation. For each measure used in the analysis, lower values indicate better measured quality.

We group the selected measures into two categories. The first category consists of routine-sensitive process measures: catheter use, anti-psychotic medication use, and anti-anxiety or hypnotic medication use. Catheter use measures the percentage of long-stay residents with a catheter inserted and left in their bladder. Catheters may be clinically appropriate in some cases, such as urinary retention, selected wound-care situations, or end-of-life care, but unnecessary catheter use can increase infection risk and may reflect facility routines around toileting assistance and monitoring. Anti-psychotic medication use measures the percentage of long-stay residents who received an anti-psychotic medication and anti-anxiety or hypnotic medication use measures the percentage of long-stay residents who received an anti-anxiety or hypnotic medication. Both of these metrics capture medication-management practices among long-stay residents. These medications may be clinically appropriate for some residents, but their use can also reflect prescribing routines, behavioral-management practices, staff monitoring, and reliance on pharmacological approaches to resident care.

The second category consists of resident outcome measures: pressure injuries, falls with major injury, weight loss, decline in physical functioning, and urinary tract infections. These measures are more directly related to resident health and functioning and may be affected by staffing-intensive care processes. Pressure injuries may reflect repositioning, skin checks, hygiene, nutrition, and monitoring. Falls with major injury capture serious resident safety events that may be affected by supervision, mobility assistance, medication management, and environmental safety. Weight loss captures clinically meaningful unintended weight loss [1] and may reflect nutrition, feeding assistance, monitoring, or changes in resident condition. Decline in physical functioning captures worsening dependence in mobility or activities of daily living. Urinary tract infections may reflect resident frailty and clinical need, but also catheter use, hydration, toileting assistance, hygiene, monitoring, and documentation practices.

2.6 Other Controls

To account for facility characteristics that may be correlated with staffing, quality, and ownership changes, we included a number of control variables that are consistent with other studies (cite here). The following variables come from NHC and HCRIS and are used as controls in all specifications: Ownership type (government, non-profit, and for-profit), chain affiliation, total number of beds, occupancy rate, payer mix (percent of revenue coming from Medicare, and percent of revenue coming from Medicaid), and acuity. Our measure of acuity is case-mix provided by CMS in the NHC archive data. During our study period CMS altered the methodology used to construct case-mix indices. As a result, raw case-mix values are

[1] This does not include residents who are on a physician-prescribed weight-loss plan.

not directly comparable over time. To address this issue, we measure acuity using relative rankings rather than levels. Specifically, we construct case-mix quartiles within each state and calendar month.

2.7 Analytic Approach

Ownership changes occur at different dates across facilities during the study period. Because these changes are not randomly assigned, our empirical strategy does not compare facilities that change ownership to facilities that do not in levels. Instead, we exploit the timing of verified control-changing ownership transitions to compare changes within a facility before and after ownership change, relative to contemporaneous changes among facilities that have not yet experienced a change or never experience one during the study period. All specifications include facility fixed effects, time fixed effects, and observable time-varying facility characteristics.

This staggered timing raises a standard concern for two-way fixed effects designs. In staggered-adoption settings, conventional two-way fixed effects estimates can be sensitive to treatment-effect heterogeneity and comparisons involving already-treated units (Goodman-Bacon, 2021). We therefore emphasize event-study estimates using alternative comparison groups. This design is well suited to the institutional setting. Following an ownership change, the facility remains in the same location, operates under the same certification framework, and serves the same local market. As a result, any post-transition changes are likely to operate through internal production decisions, including staffing, routines, and management practices, rather than through a mechanical change in local market structure.

A complication in defining event time is that the recorded ownership change month may not coincide with the beginning of the transfer process. The ownership change date we observe in HCRIS is the effective date of the change, or the closing date. However, the sale process begins earlier, when the buyer and seller negotiate and sign a contract. Ideally, we would observe the contract signing date or the month in which transition planning begins.

Because these dates are not observed, the months immediately preceding the recorded closing date may already reflect operational responses to the pending transfer. During this pre-transfer period, incumbent managers or prospective owners may adjust staffing policies, scheduling, hiring, agency use, or other operating routines. Because staffing can adjust relatively quickly, these months may not represent untreated facility behavior. For this reason, the preferred staffing event-study specification excludes $\tau = -3, -2, -1$ and uses $\tau = -4$ as the reference period. For quality outcomes, which are measured quarterly, the timing concern is different. The ownership-change quarter, $\tau = 0$, may contain care, assessment, and documentation from both before and after the transfer. We therefore exclude $\tau = 0$ and use $\tau = -1$ as the reference period.

I originally only introduced this after the specifications, but I moved it earlier this time

We estimate the following set of event-study regressions:

$$S_{im} = \sum_{\tau=-24, \tau \neq -4,-3,-2,-1}^{24} \beta_\tau \mathbf{1}(T_{im} = \tau) + X_{im}\gamma + \mu_i + \lambda_m + \varepsilon_{im}. \tag{2}$$

For staffing outcomes, S_{im} denotes RN, LPN, CNA, or total HPRD for facility i in month m. The variables $\mathbf{1}(T_{im} = \tau)$ are event-time indicators for whether facility i is τ months from ownership change. The coefficients β_τ trace monthly staffing dynamics relative to four months before the ownership change. The months $\tau = -3, -2, -1$ are excluded to remove the period most likely to be influenced by the pre-transfer adjustment window.

$$Q_{iq} = \sum_{\tau=-8, \tau \neq -1,0}^{8} \theta_\tau \mathbf{1}(T_{iq} = \tau) + X_{iq}\pi + \mu_i + \lambda_q + \nu_{iq}. \tag{3}$$

For quality outcomes, Q_{iq} denotes a selected CMS quality measure for facility i in quarter q. The variables $\mathbf{1}(T_{iq} = \tau)$ are event-time indicators for whether facility i is τ quarters from ownership change. The coefficients θ_τ trace quarterly quality dynamics relative to the quarter before ownership change. The ownership-change quarter, $\tau = 0$, is excluded because it may contain both pre- and post-transfer care, documentation, and reporting.

In both equations, X_{im} and X_{iq} denote vectors of facility controls including: ownership type, chain affiliation, beds, occupancy rate, payer mix, and state-period case-mix quartile indicators. Facility fixed effects μ_i absorb time-invariant differences across nursing homes, while month fixed effects λ_m and quarter fixed effects λ_q absorb common shocks over time. Standard errors are two-way clustered by facility and calendar period.

For quality outcomes, we estimate specifications both with and without RN, LPN, and CNA HPRD controls. Because staffing is itself affected by ownership change, the staffing-controlled specifications are not interpreted as preferred estimates of the total effect of ownership change on quality. Instead, they are used descriptively to assess whether the estimated quality responses are attenuated after conditioning on measured staffing inputs.

We also estimate average post-change specifications that replace the event-time indicators with a single post-ownership-change indicator:

$$Z_{ip} = \delta Post_{ip} + X_{ip}\gamma + \mu_i + \lambda_p + \varepsilon_{ip}. \tag{4}$$

In equation (4) $p = m$ for staffing outcomes and $p = q$ for quality outcomes. Z_{ip} denotes either a staffing outcome in facility-month im or a quality outcome in facility-quarter iq. The indicator $Post_{ip}$ equals one after the ownership change and zero otherwise. The coefficient δ summarizes the average post-ownership-change difference in the outcome, relative to the pre-change period, after accounting for facility fixed effects, time fixed effects, and time-varying controls. For staffing outcomes, the preferred post specifications are estimated after excluding the same immediate pre-transfer months used in the event-study specification. For quality outcomes, the preferred post specifications exclude the ownership-change quarter. These post specifications are used to summarize the average post-change difference and should be interpreted alongside the event-study estimates.

For staffing outcomes, we estimate event study and conventional difference-in-differences in both levels and logs. Log specifications allow coefficients to be interpreted as approximate percentage changes and reduce sensitivity to differences in baseline staffing levels across facilities. Log specifications are omitted when staffing is zero.

2.7.1 Parallel Trends

The identifying assumption of equations 2 and 3 is, absent an ownership change, treated facilities would have followed similar staffing and quality trends as not-yet-treated and untreated facilities. We assess this assumption using the pre-treatment coefficients in the event-study models and joint Wald tests of whether these coefficients are equal to zero.

For staffing outcomes, we compare pre-trend tests from the full pre-window to those from the preferred donut specification. The full pre-window includes the months immediately preceding the recorded ownership change and uses $\tau = -1$ as the reference period. The donut specification excludes $\tau \in \{-3, -2, -1\}$ and uses $\tau = -4$ as the reference period. Table 1 reports the results of these Wald tests. The improvement in the pre-trend tests after removing the immediate pre-transition months supports the interpretation that those months are influenced by the pre-transfer adjustment window.

Table 1: Joint Wald Tests of Pre-trends for Monthly Staffing

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: HPRD				
2 Year Full Pre-Window	93.04 (23)	39.36 (23)	36.25 (23)	48.73 (23)
	[0.0000]	[0.0181]	[0.0389]	[0.0013]
2 Year Window with Donut	27.52 (20)	34.10 (20)	11.36 (20)	23.90 (20)
	[0.1213]	[0.0254]	[0.9362]	[0.2466]
1 Year Window with Donut	11.23 (8)	6.39 (8)	1.43 (8)	3.73 (8)
	[0.1892]	[0.6039]	[0.9938]	[0.8806]
Panel B: Log(HPRD)				
2 Year Full Pre-Window	47.06 (23)	38.05 (23)	35.80 (23)	44.39 (23)
	[0.0022]	[0.0252]	[0.0433]	[0.0047]
2 Year Window with Donut	26.11 (20)	37.18 (20)	10.86 (20)	22.71 (20)
	[0.1622]	[0.0111]	[0.9498]	[0.3033]
1 Year Window with Donut	4.30 (8)	6.81 (8)	1.74 (8)	4.45 (8)
	[0.8294]	[0.5574]	[0.9879]	[0.8140]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.
Tested windows and reference periods: 2 Year Full Pre-Window tests $\tau = -24$ to $\tau = -2$ with reference $\tau = -1$; 2 Year Window with Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$); 1 Year Window with Donut tests $\tau = -12$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).
Sample sizes (rows): 2 Year Full Pre-Window ($N = 630, 420$); 2 Year Window with Donut ($N = 626, 311$); 1 Year Window with Donut ($N = 626, 311$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.

For quality outcomes, we conduct analogous Wald tests using the quarterly event-study specification. The full pre-window tests whether the coefficients for quarters $\tau = -8$ through $\tau = -1$ are jointly equal to zero. Our preferred quality specification excludes the ownership-change quarter, $\tau = 0$, because this quarter may combine pre- and post-transfer care, assessment, and documentation. Table 2 reports the results of these Wald tests.

Table 2: Joint Wald Tests of Pre-Trends for Quarterly Quality Measures

Outcome	**2 Year Full Pre-Window**	**2 Year, with Donut**	**1 Year, with Donut**
Catheter	12.10 (7) [0.0972]	13.00 (7) [0.0721]	2.01 (3) [0.5707]
Antipsychotic	7.49 (7) [0.3793]	7.89 (7) [0.3425]	3.43 (3) [0.3301]
Hypnotics	3.29 (7) [0.8570]	3.33 (7) [0.8532]	0.53 (3) [0.9117]
Pressure injuries	12.76 (7) [0.0781]	12.62 (7) [0.0820]	1.77 (3) [0.6211]
Falls	4.13 (7) [0.7649]	3.88 (7) [0.7937]	0.24 (3) [0.9711]
Weight loss	8.09 (7) [0.3251]	7.82 (7) [0.3489]	3.60 (3) [0.3083]
ADL increase	13.10 (7) [0.0697]	12.68 (7) [0.0802]	2.21 (3) [0.5306]
UTI	5.98 (7) [0.5422]	5.95 (7) [0.5450]	0.97 (3) [0.8081]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.
The 2 Year Full Pre-Window tests $\tau = -8$ through $\tau = -1$, with $\tau = -1$ as the omitted reference period.
The ownership-change-quarter-excluded specifications omit $\tau = 0$ because the quarter of ownership change may combine pre- and post-transfer care, assessment, and documentation.
The 1 Year Window tests $\tau = -4$ through $\tau = -1$, with $\tau = -1$ as the omitted reference period.
Sample sizes by outcome: 2 Year Full Pre-Window [Catheter=188,762; Antipsychotic=201,403; Hypnotics=150,667; Pressure injuries=140,715; Falls=197,246; Weight loss=198,690; ADL increase=200,578; UTI=196,502].
Sample sizes by outcome: 2 Year, with Donut [Catheter=187,542; Antipsychotic=200,132; Hypnotics=149,711; Pressure injuries=139,764; Falls=195,998; Weight loss=197,438; ADL increase=199,311; UTI=195,249].
Sample sizes by outcome: 1 Year, with Donut [Catheter=187,542; Antipsychotic=200,132; Hypnotics=149,711; Pressure injuries=139,764; Falls=195,998; Weight loss=197,438; ADL increase=199,311; UTI=195,249].
All specifications include facility and quarter fixed effects and covariates: government + non_profit + chain + beds + occupancy_rate + pct_medicare + pct_medicaid + cm_q_state_2 + cm_q_state_3 + cm_q_state_4.

2.7.2 Additional Analyses

3 Results

3.1 Summary Statistics

Table 3 reports summary statistics for the monthly staffing sample. The final sample consists of approximately 7,806 unique skilled nursing facilities and 632,231 facility-month observations spanning January 2017 through June 2024. Of these facilities, 1589 experience an ownership change during the sample period.

Table 3: Summary Statistics

Variable	Mean	SD
Panel A: Outcome variables		
RN HPRD	0.427	0.309
LPN HPRD	0.796	0.325
CNA HPRD	2.100	0.548
Total HPRD	3.324	0.793
Panel B: Control variables		
Government (dummy)	0.101	0.301
Non-profit (dummy)	0.171	0.377
Chain affiliation (dummy)	0.546	0.498
Beds	108.4	57.9
Occupancy rate (%)	76.4	15.6
% Medicare	13.0	11.1
% Medicaid	60.6	20.3
Acuity quartile 2 (state-month)	0.231	0.422
Acuity quartile 3 (state-month)	0.230	0.421
Acuity quartile 4 (state-month)	0.233	0.423

Notes: The unit of observation is facility–month. HPRD denotes hours per resident day. RN, LPN, and CNA denote registered nurses, licensed practical nurses, and certified nursing assistants, respectively. Total HPRD is the sum of RN, LPN, and CNA HPRD. Occupancy rate is the ratio of residents to certified beds (percent). % Medicare and % Medicaid are the shares of residents covered by Medicare and Medicaid, respectively. Government and Non-profit are ownership-type indicator variables (for-profit omitted category). Chain affiliation is an indicator for membership in a multi-facility chain. Acuity quartiles are state–month quartiles of resident acuity (case-mix) with quartile 1 omitted. Rows = 632,231; Facilities = 7,806; Treated facilities = 1,589; Period = 2017/01–2024/06; Average months per facility = 81.0.

Table 4 reports summary statistics for the quarterly quality sample. The final sample consists of x unique skilled nursing facilities and x facility-quarter observations over the same period as the staffing sample. Of this sample, x go through an ownership change during the sample period.

Table 4: Summary Statistics

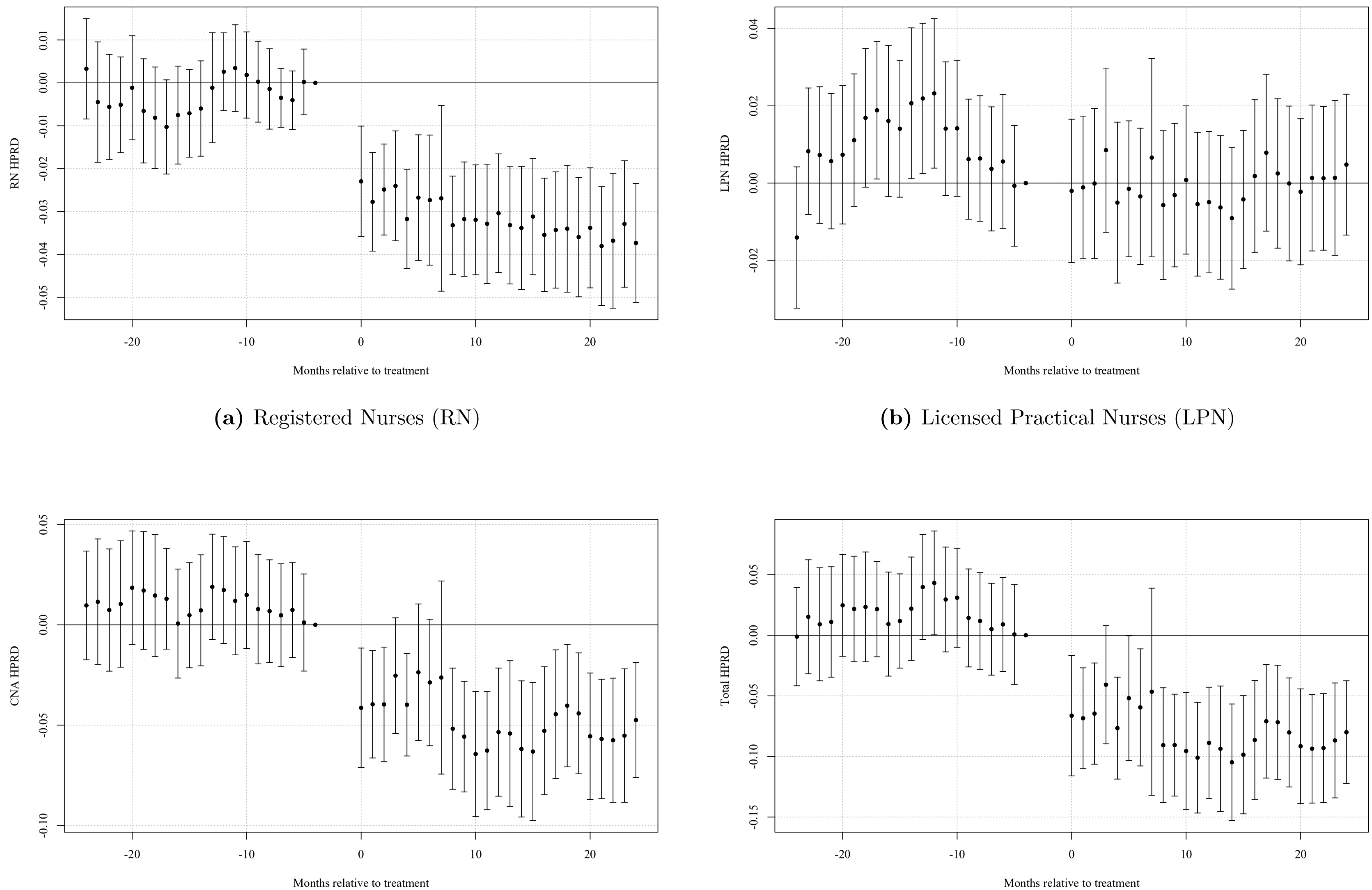
Variable	Mean	SD
Panel A: Selected quality metrics		
Catheter	1.586	2.121
Antipsychotic	14.306	9.287
Hypnotics	20.394	10.269
Pressure injuries	7.793	5.259
Falls with major injury	3.384	2.838
Weight Loss	6.229	4.864
ADL Increase	15.001	8.801
Urinary Tract Infections	2.486	3.237
Panel B: Staffing and control variables		
RN HPRD	0.422	0.299
LPN HPRD	0.795	0.316
CNA HPRD	2.092	0.533
Government (dummy)	0.101	0.301
Non-profit (dummy)	0.170	0.376
Chain affiliation (dummy)	0.549	0.498
Beds	109.0	57.8
Occupancy rate (%)	76.6	15.4
% Medicare	13.3	12.3
% Medicaid	60.7	20.4
Acuity quartile 2 (state-quarter)	0.249	0.433
Acuity quartile 3 (state-quarter)	0.248	0.432
Acuity quartile 4 (state-quarter)	0.252	0.434

Notes: The unit of observation is facility–quarter. RN, LPN, and CNA denote registered nurses, licensed practical nurses, and certified nursing assistants, respectively. HPRD denotes hours per resident day. Occupancy rate is the ratio of residents to certified beds (percent). % Medicare and % Medicaid are the shares of residents covered by Medicare and Medicaid, respectively. Government and Non-profit are ownership-type indicator variables (for-profit omitted category). Chain affiliation is an indicator for membership in a multi-facility chain. Acuity quartiles are state–quarter quartiles of resident acuity (case-mix) with quartile 1 omitted. Rows = 208,657; Facilities = 7,768; Treated facilities = 1,588; Period = 2017Q1–2024Q2; Average quarters per facility = 26.9.

3.2 Nurse Staffing Results

Figure 1 presents event study estimates of the effect of ownership changes on nursing staffing outcomes (i.e., Equation 2). Following an ownership transition, staffing declines persist for many periods, with the largest and most precise effects occurring for registered nurses (RN), certified nursing assistants (CNA), and total nursing hours per resident day.

RN staffing declines immediately following ownership changes and remains far below pre-transition levels throughout the post-treatment period. CNA staffing exhibits a similar but slightly smaller decline, while total nursing hours mirror the combined reductions across staff types. In contrast, LPN staffing shows limited and statistically insignificant changes over time. These dynamic patterns indicate that ownership transitions lead to sustained reductions in staffing rather than short run adjustments.



(a) Registered Nurses (RN)

(b) Licensed Practical Nurses (LPN)

(c) Certified Nursing Assistants (CNA)

(d) Total Nursing Hours

Figure 1: Dynamic Effects of Ownership Change on Nursing Staffing. Event study estimates of the effect of a change in ownership on nursing staff levels in hours per resident day (HPRD). Points show coefficient estimates relative to the reference period $\tau = -4$; vertical bars denote 95% confidence intervals. The three months prior to the ownership change ($\tau = -3, -2, -1$) are excluded. All specifications include control variables and facility and month fixed effects.

To summarize these dynamic effects with a single average treatment estimate, Table 5 reports two-way fixed effects estimates of the ownership change effect (i.e., Equation 4). Panel A reports results in levels, while Panel B reports logged results. Consistent with the event study results, ownership changes lead to meaningful reductions in staffing levels. RN staffing declines by approximately 0.03 hours per resident day (roughly 9 percent), CNA staffing falls by about 0.05 hours (approximately 3 percent), and total staffing decreases by 0.08 hours per resident day (roughly 3 percent). LPN staffing shows no statistically significant change. Estimates are nearly identical whether the observations for the three month period before the ownership change occurred is included or excluded from the model.

Table 5: Two-Way Fixed Effects Estimates of *post* on Staffing Outcomes (Baseline, Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
HPRD	-0.031^{***}	-0.000	-0.052^{***}	-0.083^{***}
	(0.005)	(0.006)	(0.009)	(0.012)
Log(HPRD)	-0.088^{***}	0.003	-0.026^{***}	-0.025^{***}
	(0.016)	(0.009)	(0.005)	(0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses. The table reports staffing levels (HPRD) and log staffing levels (Log(HPRD)). Sample: Without anticipation ($N_{\text{HPRD}} = 626,311$; $N_{\text{Log}} = 626,311$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

3.2.1 Effects Before and During Covid-19 Pandemic

Table 6 examines whether these effects differ before and after the COVID-19 pandemic. The magnitude of the decline is roughly halved in the post pandemic period for RN, CNA, and total HPRD, suggesting that pandemic induced shifts in labor markets and staffing substantially reduced the impact of ownership transitions.

Table 6: TWFE Estimates of *post*: Pre- vs Post-pandemic Periods (Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Staffing Levels in HPRD				
Pre-Pandemic Period (2017/01 - 2019/12)	-0.029^{***}	-0.019^{**}	-0.063^{***}	-0.111^{***}
	(0.006)	(0.008)	(0.014)	(0.019)
Pandemic Period (2020/04 - 2024/06)	-0.018^{**}	0.000	-0.041^{***}	-0.058^{***}
	(0.007)	(0.008)	(0.015)	(0.020)
Panel B: Log Staffing Levels in HPRD				
Pre-Pandemic Period (2017/01 - 2019/12)	-0.073^{***}	-0.014	-0.032^{***}	-0.034^{***}
	(0.024)	(0.012)	(0.007)	(0.006)
Pandemic Period (2020/04 - 2024/06)	-0.065^{***}	0.001	-0.020^{**}	-0.019^{***}
	(0.021)	(0.011)	(0.008)	(0.006)

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses. Panel A reports levels (HPRD); Panel B reports logs (HPRD). Sample sizes: Row 1 ($N_{\text{levels}} = 252,453$; $N_{\text{logs}} = 252,453$), Row 2 ($N_{\text{levels}} = 356,174$; $N_{\text{logs}} = 356,174$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.
Pre-pandemic 2017/01–2019/12; Pandemic 2020/04–2024/06.

Overall, ownership changes lead to sizable and persistent reductions in nursing staffing, with effects concentrated among RN, CNA, and total hours per resident day. These reductions are larger prior to the COVID-19 pandemic and among facilities that were not part of a chain at baseline. Together, the event study and TWFE results indicate that ownership transitions systematically worsen staffing conditions in nursing homes rather than inducing temporary adjustments.

3.3 Quality Measure Results

[Insert quality metric results here] [Do we want to present the event study results? Are we only presenting DiD results?] [Change LaTeX table output to be an input and not standalone doc*]

3.4 Robustness

This section presents a series of robustness checks designed to assess the validity of our identifying assumptions and the sensitivity of our results to alternative model choices. Across all exercises, the estimated effects of ownership change on staffing remain stable in sign, magnitude, and statistical significance. Most of these will have tables/plots that will go in the appendix.

3.4.1 Assumptions

Our main event-study specification is estimated using two-way fixed effects (TWFE) with staggered treatment timing. In this setting, TWFE can implicitly use already-treated facilities as controls for cohorts treated later, which may distort event-study estimates when treatment effects are dynamic or heterogeneous across cohorts. To assess sensitivity to this issue, we re-estimate the event study using a stacked difference-in-differences design.

In the stacked approach, we construct cohort-specific samples for each treatment month g. For a given cohort, treated facilities are those with an ownership change in month g, and the control group consists only of facilities that are never treated or not-yet-treated relative to g (i.e., with treatment dates after g). We restrict each cohort-specific sample to a symmetric event window around g and then stack these cohort datasets into a single analysis file. Estimation includes the same covariates as the baseline specification and fixed effects that absorb facility-level time-invariant heterogeneity and common calendar shocks; standard errors are clustered at the facility level.

Appendix Figures 2 through 5 show that the stacked event-study estimates closely track the baseline TWFE event-study results: the post-transition dynamics and the relative magnitudes across staffing types remain qualitatively unchanged. In addition, Appendix Table 12 reports joint Wald tests of pre-treatment coefficients in the stacked specification (under the donut design used in the baseline), which are consistent with flat pre-trends in the remaining pre-period. Together, these results suggest that the main findings are not driven by treated-as-control comparisons that can arise in TWFE under staggered adoption.

3.4.2 Event Window and "Donut"

As we have previously discussed, because of changes to staffing in anticipation of a formal ownership change, the assumptions of parallel trends is violated. As a result of this, in our baseline specifications we "donut" equation **??** and equation 4 by removing $\tau \in \{-3, -2, -1\}$. To test the validity of removing this set of pre-treatment coefficients, we estimate equations **??** and 4 with different "donut" sets and show that the estimated post-treatment effects remain largely unchanged. Row 6 of Table 11 reports estimates of equation 4 with a wider anticipatory period excluded from the sample. The results from row 6 show no differences in values or statistical significance. Similarly, row 7 reports estimates under a smaller anticipatory window that is excluded from the data, and we see no differences in the results.

3.4.3 Gap Size

Because staffing data is occasionally missing or reported with gaps, we examine whether our results are sensitive to the length of time between consecutive observations. Restricting the sample to facilities with smaller reporting gap shows that estimates are similar to those in the full sample, indicating that data gaps do not drive our findings. Row 2 of Table 11 reports our results for equation 4 with the restriction that facilities with a reporting gap of larger than 6 are excluded from our sample. Compared to the baseline (row 1) this result is identical across staff types and statistical significance. The exact same result is true for row 3 which reports estimates for equation 4 where facilities with a reporting gap of greater than 3 are removed. Row 4 and 5 report estimates where gap greater than 1 or equal to 0 respectively, are removed. Similarly, we see no differences in these results with respect to the baseline.

4 Discussion

We find that ownership changes are followed by persistent reductions in nurse staffing, particularly in RN hours, CNA hours, and total nursing hours per resident day. In a regulated sector with limited short-run pricing flexibility, staffing is one of the most immediately adjustable margins, so changes in staffing provide a useful window into how new owners reshape internal production decisions. Our event study estimates, together with the results across alternative specifications, suggest that these are not short-lived reductions around the recorded ownership change month, but sustained adjustments following a change in control.

4.1 No declines in LPN hours

One notable result is that LPN staffing shows little to no decline following ownership change, even as RN and CNA hours fall. This pattern is consistent with substitution across nurse types arising from differences in training, scope of practice, and how tasks are organized within nursing homes. RNs perform higher-skill clinical and supervisory functions, while CNAs provide the majority of hands-on daily care (Lin, 2014). LPNs occupy an intermediate role where they can perform a range of routine clinical tasks that overlap with RN responsibilities, while also helping maintain day-to-day coverage in ways that partly offset reductions in CNA time.

This makes LPNs a plausible way to adjust when new owners seek to reduce labor costs without generating large immediate changes in service provision. CNAs generally cannot substitute for RNs on licensed or supervisory tasks, and RNs are not typically used to replace CNA-level direct-care work because of wage differences and the organization of production. LPNs, by contrast, can absorb some routine clinical work that might otherwise require RN time, while also supporting continuity of care when CNA hours are reduced. In that sense, the relative stability of LPN staffing is consistent with compositional adjustment rather than uniform staffing cuts.

4.2 Are the staffing reductions economically meaningful?

An important question is whether the estimated staffing reductions are large in economic terms. Benchmarking the estimates against baseline staffing levels helps provide context. In our sample, mean staffing levels are 0.427 RN HPRD, 0.796 LPN HPRD, 2.100 CNA HPRD, and 3.324 total HPRD. In the baseline TWFE specification, an ownership change reduces RN staffing by 0.032 HPRD, CNA staffing by 0.054 HPRD, and total staffing by 0.086 HPRD, while LPN staffing does not change significantly.

These effects correspond to declines of roughly 7.5% for RN staffing, 2.6% for CNA staffing, and 2.6% for total staffing. Expressed in more intuitive terms, 0.032 RN HPRD is about 1.9 fewer RN minutes per resident-day, 0.054 CNA HPRD is about 3.2 fewer CNA minutes per resident-day, and 0.086 total HPRD is about 5.2 fewer total nursing minutes per resident-day.

Our estimates should not be interpreted as implying large welfare effects simply because they are statistically significant. At the same time, staffing is a core input into care production, so even modest staffing changes could still matter, especially in facilities operating close to staffing constraints. In larger facilities, some reductions may be diffused across many residents and workers; in smaller facilities, comparable average reductions may be more likely to bind and produce noticeable changes in quality. Thus, the economic significance of these estimates for resident care is ultimately ambiguous in our paper. What we can show is that ownership changes affect staffing inputs; whether those input changes translate into meaningful changes in deficiencies, hospitalizations, mortality, or other outcomes remains an open empirical question.

4.3 Heterogeneity

Two heterogeneity patterns provide additional insight into mechanisms. First, staffing declines are larger for non-chain facilities than for chain-affiliated facilities. This pattern is consistent with organizational buffers and internal labor market capacity in chains: chain facilities may have greater ability to stabilize staffing, share managerial practices, and respond to financial or operational shocks without reducing measured staffing as sharply. In contrast, independent facilities may face tighter constraints and fewer margin-stabilizing tools, making staffing a more immediate adjustment channel. Second, estimated staffing reductions are smaller during the COVID-19 period than in the pre-pandemic period. A natural interpretation is that pandemic-era labor market tightness limited owners' ability to implement further staffing reductions, while heightened scrutiny and operational disruptions may have constrained the feasible set of staffing adjustments.

4.4 Implications

A key contribution of this paper is to establish staffing as an adjustment margin following ownership changes, even in a setting where prices are largely fixed and production decisions are heavily regulated. However, staffing is an intermediate input, and whether these ownership-induced staffing reductions reflect efficiency gains (e.g., reduced slack, improved task allocation) or quality-reducing cost containment is ultimately an empirical question about outcomes. A natural extension is to link ownership transitions to resident-centered measures (e.g., deficiencies, rehospitalizations, mortality, or other quality indicators) and test whether the staffing reductions documented here coincide with changes in measured quality. Such analysis could also help with the interpretation of the LPN substitution mechanism: if RN and CNA hours fall while quality is unchanged, this would be more consistent with task reallocation and efficiency; if quality deteriorates, it would indicate that staffing reductions likely reflect quality-reducing cost containment. The results here motivate a broader evaluation of ownership changes that jointly considers staffing composition and resident outcomes.

4.5 Limitations

First, despite the two-step approach to identifying ownership transitions and the use of a donut window to address anticipatory adjustments, the timing of ownership change may still be measured with error, and some operational changes may precede the recorded transition date. Second, while the event-study design with fixed effects and pre-trend diagnostics helps address confounding, ownership changes may coincide with unobserved shocks (e.g., financial distress) that independently affect staffing trajectories. Third, PBJ-based staffing measures may be affected by reporting completeness; although robustness checks suggest the findings are not driven by reporting gaps, measurement error remains possible. Finally, because the analysis focuses on staffing inputs, the paper does not directly estimate effects on resident outcomes; interpreting welfare implications therefore requires either stronger assumptions about the staffing–quality relationship or additional outcome analyses.

5 Conclusion

Ownership changes are common in U.S. nursing homes, yet their implications for care production are not well understood. Using panel data from 2017 through 2024, we identify ownership transitions of true economic control and estimate their effects on nurse hours per resident day. We find that ownership changes lead to declines in RN staffing, CNA staffing, and total nursing hours per resident day, while LPN staffing remains stable. These effects are larger in non-chain facilities and smaller during the COVID-19 period.

Our findings suggest that ownership changes affect nursing home operations through internal production decisions, even when the set of providers in a market does not change. In a sector where short-run prices are heavily constrained, staffing is an especially important adjustment margin. At the same time, the estimated reductions are modest in absolute terms, so the welfare implications for residents cannot be determined from staffing evidence alone. The results should therefore be interpreted as evidence that ownership transitions change staffing inputs, not as definitive evidence about resident outcomes.

This paper contributes to the literature by shifting away from ownership changes solely as a market-structure event and toward its consequences for the internal organization of care. A natural next step is to link ownership transitions to resident outcomes such as deficiencies, hospitalizations, and mortality, and to study the mechanisms through which new owners alter staffing and task allocation. Doing so would help clarify when ownership transitions reflect efficiency-enhancing reorganization and when they instead reflect quality-reducing cost containment.

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Tables

Table 7: TWFE Post Estimates with Interaction: Chain vs Non-chain (Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Staffing Levels in HPPD				
Chain (baseline 2017Q1): *post*	-0.038^{***}	-0.000	-0.054^{***}	-0.092^{***}
	(0.006)	(0.007)	(0.011)	(0.015)
Difference (Non-chain − Chain): *post* × Non-chain	0.006	0.005	-0.016	-0.005
	(0.011)	(0.013)	(0.021)	(0.028)
Non-chain effect: *post* + interaction	-0.032^{***}	0.005	-0.070^{***}	-0.097^{***}
	(0.009)	(0.011)	(0.019)	(0.025)
Panel B: Log Staffing Levels				
Chain (baseline 2017Q1): *post*	-0.127^{***}	-0.003	-0.030^{***}	-0.030^{***}
	(0.020)	(0.012)	(0.006)	(0.005)
Difference (Non-chain − Chain): *post* × Non-chain	0.054	0.014	-0.003	0.003
	(0.036)	(0.021)	(0.011)	(0.009)
Non-chain effect: *post* + interaction	-0.073^{**}	0.011	-0.033^{***}	-0.027^{***}
	(0.029)	(0.017)	(0.010)	(0.008)

Notes: Each cell reports the indicated coefficient with two-way clustered standard errors (by facility and month) in parentheses.
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Sample: facilities with non-missing baseline chain status (2017Q1), without anticipation ($N = 561,440$).
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 8: TWFE Post Estimates with Interaction: Pre-pandemic vs Pandemic (Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Staffing Levels in HPPD				
Pre-pandemic: *post*	-0.023^{***}	-0.016^{**}	-0.040^{***}	-0.079^{***}
	(0.006)	(0.008)	(0.012)	(0.017)
Difference (Pandemic − Pre): *post* × Pandemic	-0.010^{*}	0.019^{***}	-0.017	-0.009
	(0.006)	(0.007)	(0.011)	(0.015)
Pandemic effect: *post* + interaction	-0.033^{***}	0.002	-0.057^{***}	-0.088^{***}
	(0.005)	(0.006)	(0.010)	(0.013)
Panel B: Log Staffing Levels				
Pre-pandemic: *post*	-0.076^{***}	-0.021^{*}	-0.014^{**}	-0.022^{***}
	(0.024)	(0.012)	(0.007)	(0.005)
Difference (Pandemic − Pre): *post* × Pandemic	-0.020	0.027^{***}	-0.016^{**}	-0.005
	(0.022)	(0.010)	(0.006)	(0.005)
Pandemic effect: *post* + interaction	-0.096^{***}	0.006	-0.030^{***}	-0.027^{***}
	(0.017)	(0.009)	(0.005)	(0.004)

Notes: Each cell reports the indicated coefficient with two-way clustered standard errors (by facility and month) in parentheses.
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Sample: 2017/01–2019/12 and 2020/04–2024/06 (excluding 2020/01–2020/03), without anticipation ($N = 610,364$).
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 9: TWFE DiD Estimates of *post* on Staffing Outcomes (Stacked Sample, Baseline Donut)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
HPPD	-0.023^{***}	-0.007	-0.051^{***}	-0.082^{***}
	(0.004)	(0.005)	(0.008)	(0.011)
Log(HPPD)	-0.066^{***}	-0.003	-0.027^{***}	-0.026^{***}
	(0.014)	(0.008)	(0.004)	(0.003)

Notes: Each cell reports the coefficient on *post* with facility-clustered standard errors in parentheses. The stacked sample includes never-treated and not-yet-treated controls for each cohort; the donut excludes $\tau \in \{-3, -2, -1\}$. Sample sizes: $N_{\text{HPPD}} = 23,877,687$; $N_{\text{Log}} = 23,492,034$.
Specifications include facility-by-cohort fixed effects (stack_id), calendar-month fixed effects, cohort fixed effects, and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 10: Two-Way Fixed Effects Estimates of *post* on Staffing Outcomes Using MCR Event Timing

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: With anticipation				
HPRD	-0.031^{***}	-0.000	-0.052^{***}	-0.083^{***}
	(0.005)	(0.006)	(0.009)	(0.012)
Log(HPRD)	-0.088^{***}	0.003	-0.026^{***}	-0.025^{***}
	(0.016)	(0.009)	(0.005)	(0.004)
Panel B: Without anticipation				
HPRD	-0.032^{***}	-0.000	-0.054^{***}	-0.086^{***}
	(0.005)	(0.006)	(0.009)	(0.013)
Log(HPRD)	-0.094^{***}	0.003	-0.028^{***}	-0.026^{***}
	(0.016)	(0.009)	(0.005)	(0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses.
Panel A uses the full baseline sample with anticipation periods included. Panel B excludes observations with *anticipation2*=1 (that is, event time $\in \{-3, -2, -1\}$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators, when available in the panel.
Estimation sample sizes for Panel A HPRD models: RN=632,227; LPN=632,227; CNA=632,227; Total=632,227.
Estimation sample sizes for Panel A log(HPRD) models: RN=626,782; LPN=627,689; CNA=632,111; Total=632,227.
Estimation sample sizes for Panel B HPRD models: RN=628,115; LPN=628,115; CNA=628,115; Total=628,115.
Estimation sample sizes for Panel B log(HPRD) models: RN=622,716; LPN=623,586; CNA=627,999; Total=628,115.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 11: TWFE estimates of *post* (without anticipation).

	N	Outcomes RN	LPN	CNA	Total
Panel A: Staffing Levels in HPPD					
(1) Baseline	628,107	-0.032^{***} (0.005)	0.000 (0.006)	-0.054^{***} (0.009)	-0.086^{***} (0.013)
(2) Sample excludes *gap* > 6	627,923	-0.032^{***} (0.005)	-0.000 (0.006)	-0.054^{***} (0.009)	-0.086^{***} (0.013)
(3) Sample excludes *gap* > 3	627,626	-0.032^{***} (0.005)	-0.000 (0.006)	-0.054^{***} (0.009)	-0.086^{***} (0.013)
(4) Sample excludes *gap* > 1	623,585	-0.032^{***} (0.005)	0.000 (0.006)	-0.055^{***} (0.009)	-0.086^{***} (0.013)
(5) Sample excludes *gap* > 0	623,585	-0.032^{***} (0.005)	0.000 (0.006)	-0.055^{***} (0.009)	-0.086^{***} (0.013)
(6) Drop $t \in \{-4, -3, -2, -1\}$	626,729	-0.032^{***} (0.005)	0.000 (0.006)	-0.055^{***} (0.009)	-0.086^{***} (0.013)
(7) Drop $t \in \{-2, -1\}$ only	628,107	-0.032^{***} (0.005)	0.000 (0.006)	-0.054^{***} (0.009)	-0.086^{***} (0.013)
Panel B: Log Staffing Levels in HPPD					
(1) Baseline	628,107	-0.092^{***} (0.016)	0.003 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(2) Sample excludes *gap* > 6	627,923	-0.092^{***} (0.016)	0.003 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(3) Sample excludes *gap* > 3	627,626	-0.092^{***} (0.016)	0.003 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(4) Sample excludes *gap* > 1	623,585	-0.092^{***} (0.016)	0.004 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(5) Sample excludes *gap* > 0	623,585	-0.092^{***} (0.016)	0.004 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(6) Drop $t \in \{-4, -3, -2, -1\}$	626,729	-0.092^{***} (0.016)	0.003 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(7) Drop $t \in \{-2, -1\}$ only	628,107	-0.092^{***} (0.016)	0.003 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)

Notes: Each cell reports the *post* coefficient with two-way clustered standard errors at the facility and month levels. Panel A uses staffing levels (HPPD); Panel B uses logs of HPPD.
All specifications include facility and month fixed effects and controls for ownership (government, non-profit, chain), beds, occupancy rate, payer mix (Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 12: Joint Wald Tests of Pre-trends (Stacked Event Study) — Levels

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
2 Year Window with Donut	28.46 (20)	35.10 (20)	20.25 (20)	24.08 (20)
	[0.0990]	[0.0196]	[0.4423]	[0.2388]
1 Year Window with Donut	5.26 (8)	24.35 (8)	6.59 (8)	15.99 (8)
	[0.7300]	[0.0020]	[0.5817]	[0.0425]

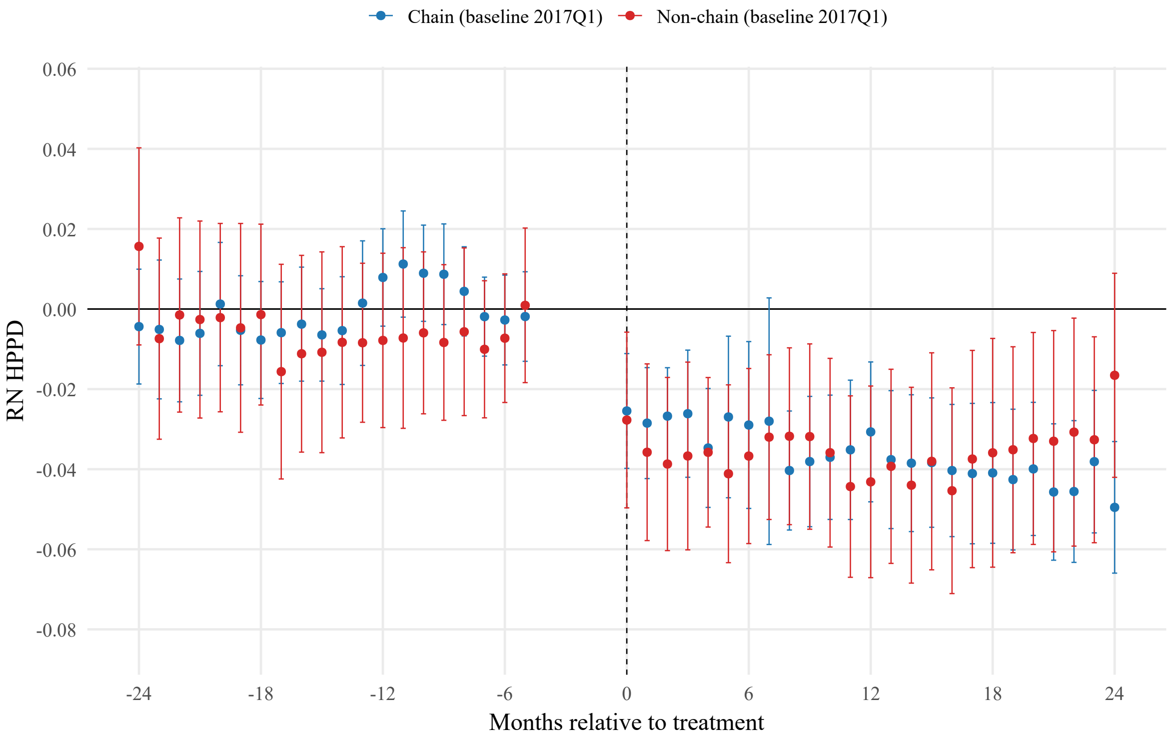
Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.

Tested windows and reference periods: 2 Year Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$); 1 Year Donut tests $\tau = -12$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).

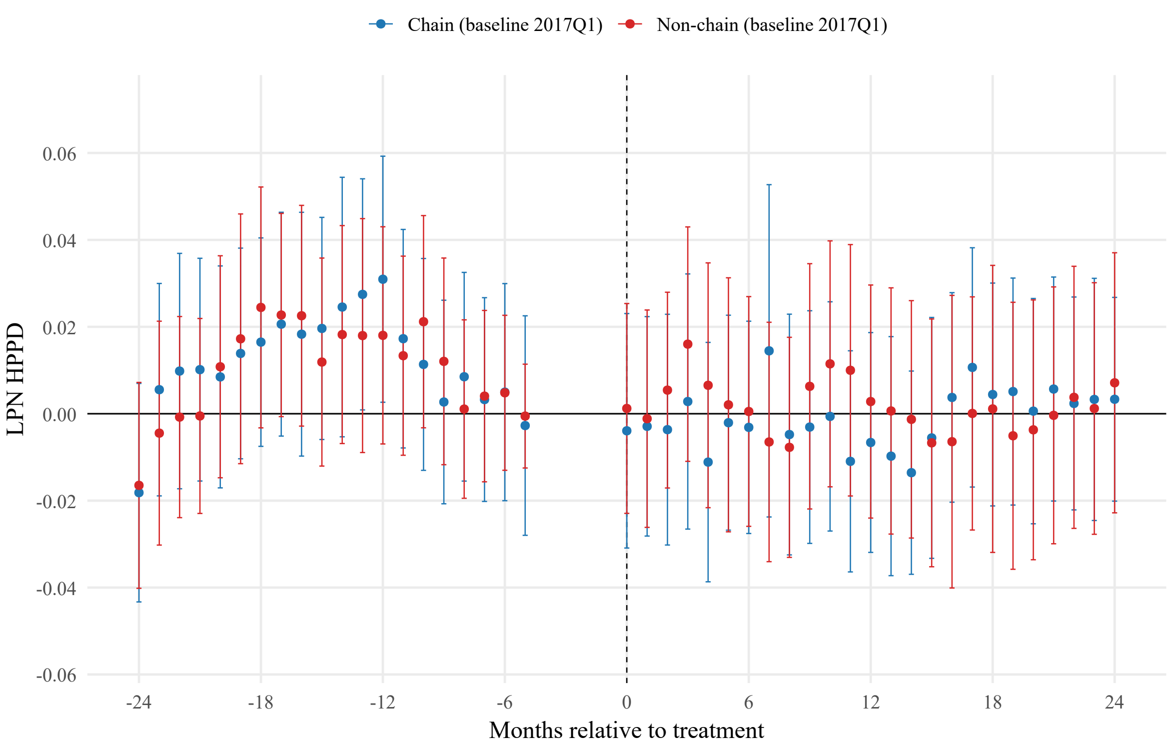
Sample sizes (stacked rows): 2 Year Donut ($N = 23,877,687$); 1 Year Donut ($N = 13,121,085$).

All specifications include stacked-unit fixed effects (facility-by-cohort), calendar-month fixed effects, cohort fixed effects, and baseline covariates. Standard errors are clustered at the facility level.

Figures



Chain (baseline 2017Q1)
Non-chain (baseline 2017Q1)
RN HPPD
0.06
0.04
0.02
0.00
-0.02
-0.04
-0.06
-0.08
-24
-18
-12
-6
0
6
12
18
24
Months relative to treatment



Chain (baseline 2017Q1)
Non-chain (baseline 2017Q1)
LPN HPPD
0.06
0.04
0.02
0.00
-0.02
-0.04
-0.06
-24
-18
-12
-6
0
6
12
18
24
Months relative to treatment

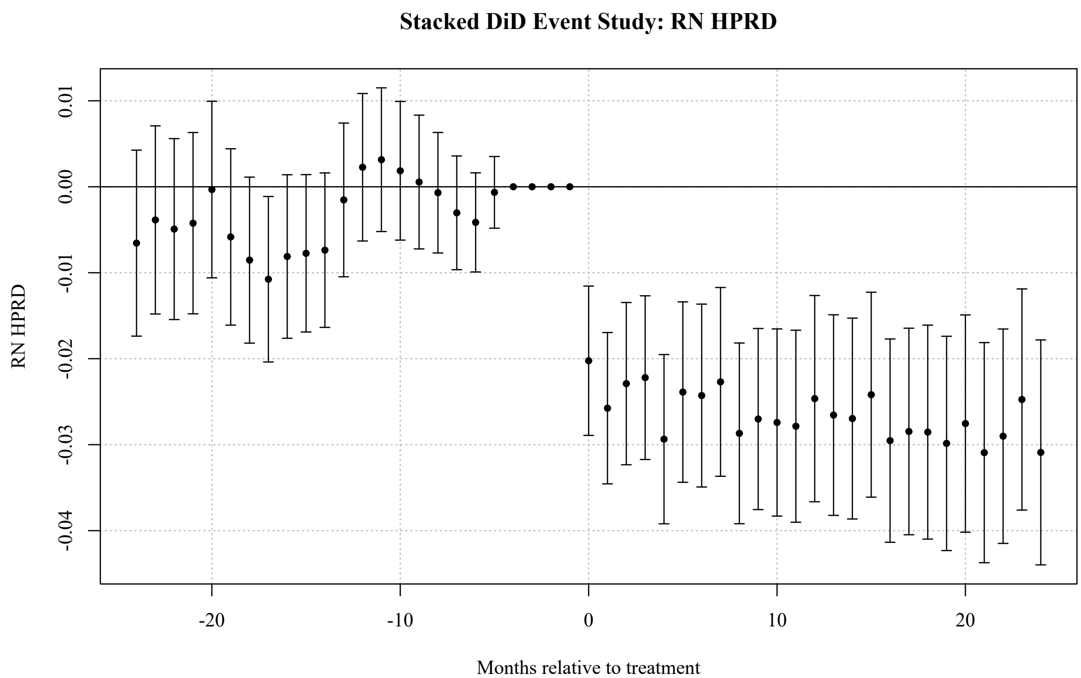


Figure 2: Stacked event-study estimates (RN staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

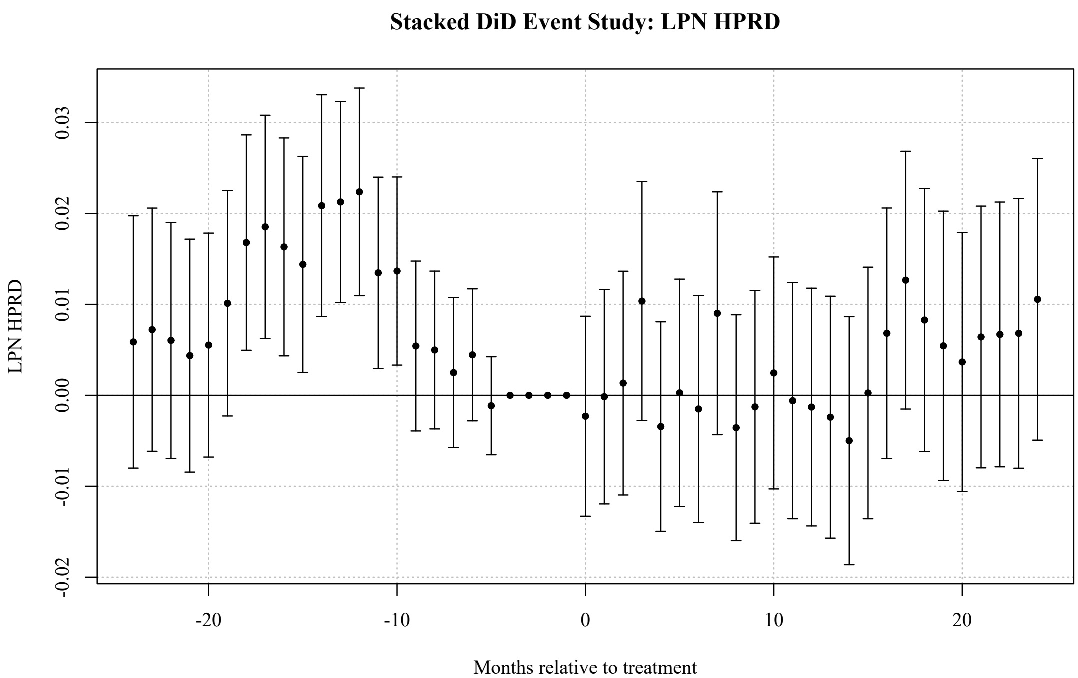


Figure 3: Stacked event-study estimates (LPN staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

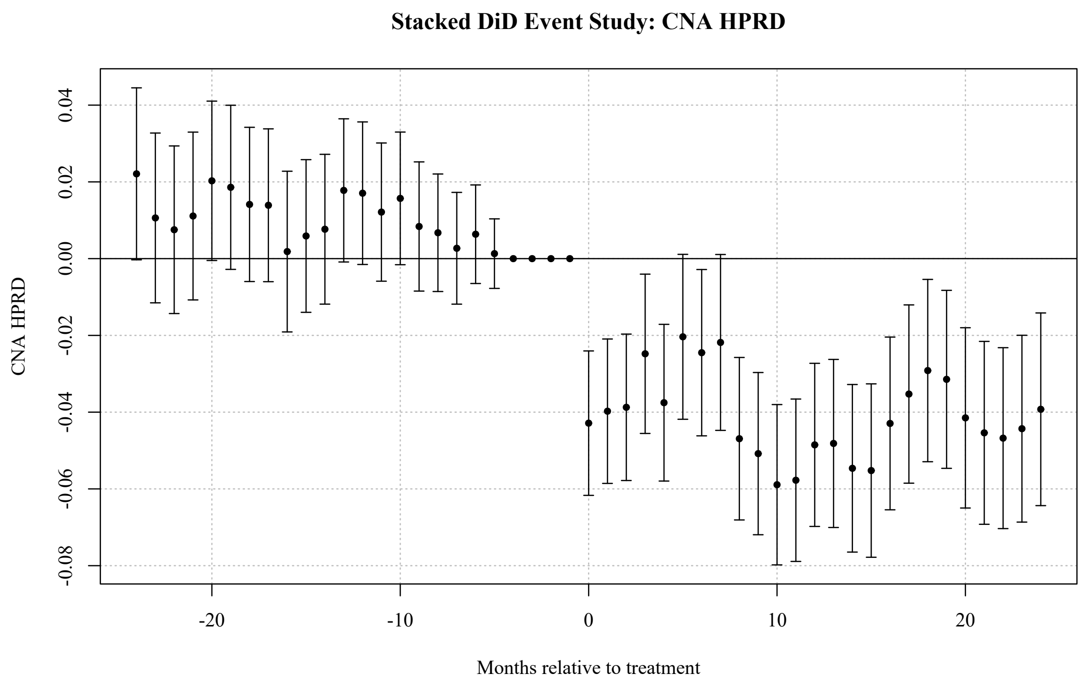


Figure 4: Stacked event-study estimates (CNA staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

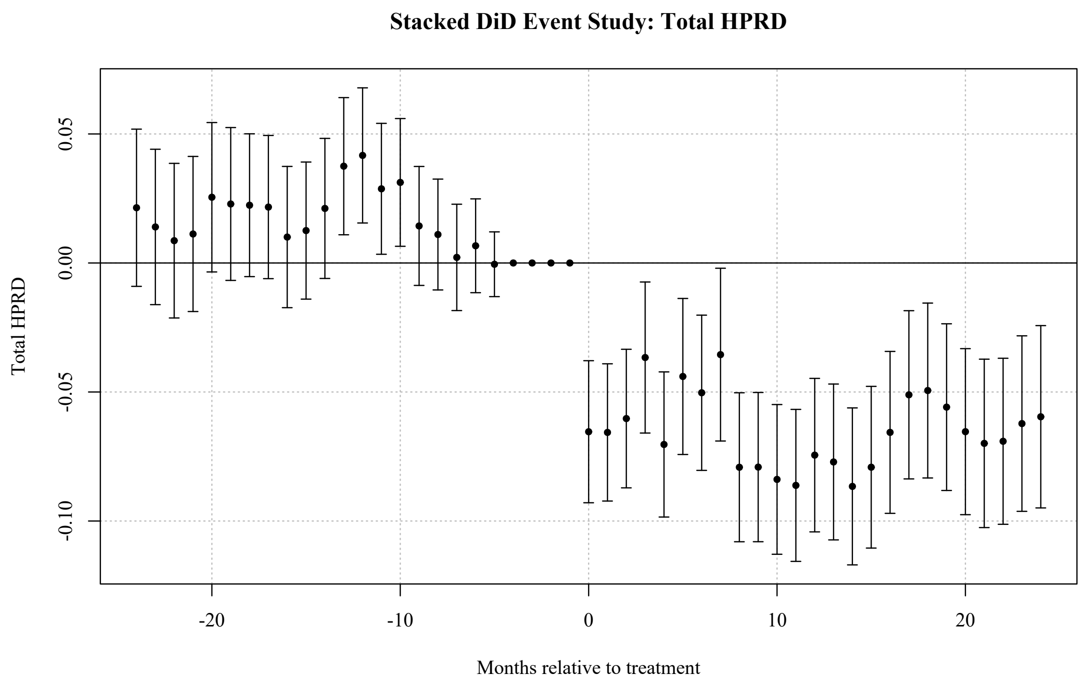


Figure 5: Stacked event-study estimates (Total staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

Appendix A. Placeholder